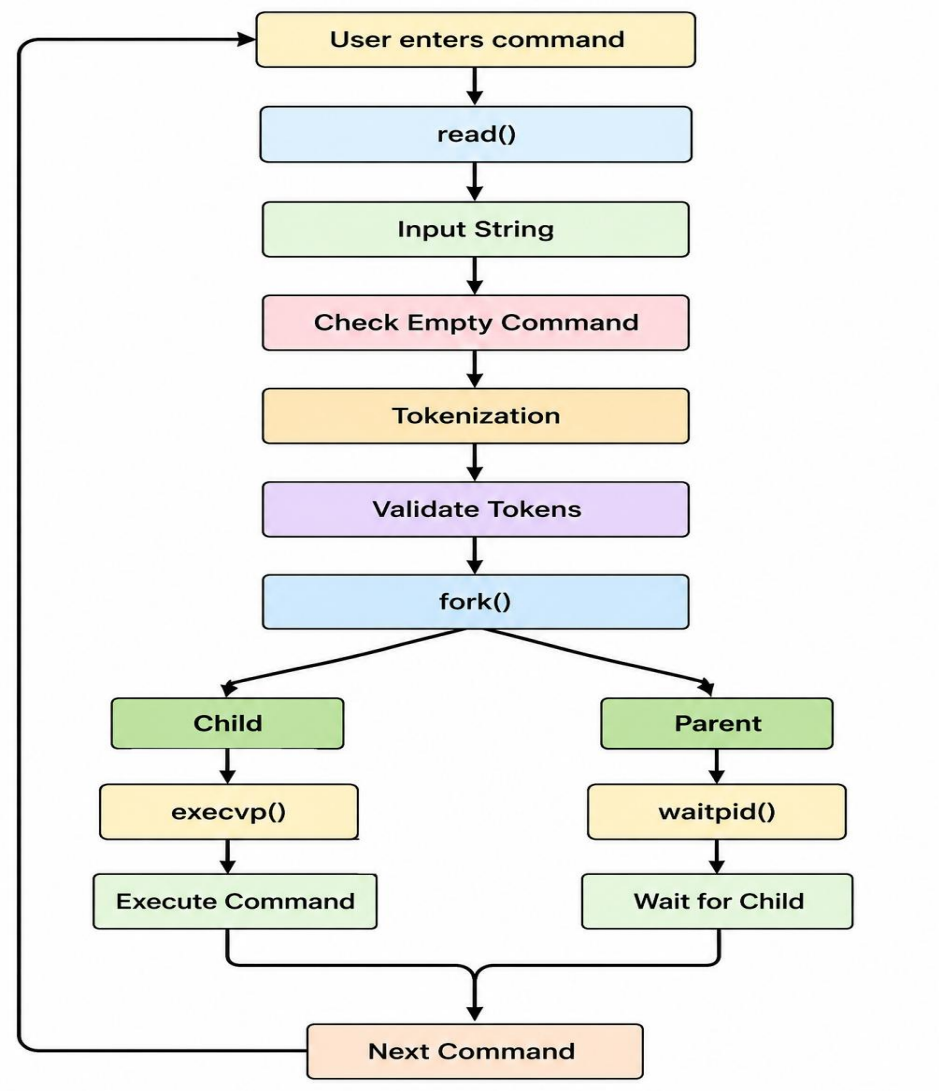
Refer the following link to understand the Parsing and Lexical analysis
https://www.geeksforgeeks.org/compiler-design/introduction-of-lexical-analysis/

Task 1:
To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output and Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures

Aim
To split a shell command into tokens and identify the individual arguments.
Program

Flow Chart:



Program

```
#include <stdio.h>
#include <string.h>

#define MAX_TOKENS 20

int main() {
    char input[200];
    char *token;
    char *tokens[MAX_TOKENS];
    int count = 0;

    printf("myShell> ");
    fgets(input, sizeof(input), stdin);

    token = strtok(input, " \t\n");

    while (token != NULL && count < MAX_TOKENS) {
        tokens[count++] = token;
        token = strtok(NULL, " \t\n");
    }

    if (count == 0) {
        printf("Empty command.\n");
        return 0;
    }

    printf("\n--- Parsed Arguments ---\n");

    for (int i = 0; i < count; i++) {
        printf("Argument %d: %s\n", i + 1, tokens[i]);
    }

    printf("Tokenization completed successfully.\n");

    return 0;
}
```

Output

```
amrutha@blue:~/OSSP$ mkdir -p SKILL_Week4
amrutha@blue:~/OSSP$ cd SKILL_Week4
amrutha@blue:~/OSSP/SKILL_Week4$ pwd
/home/amrutha/OSSP/SKILL_Week4
amrutha@blue:~/OSSP/SKILL_Week4$ nano parser.c
amrutha@blue:~/OSSP/SKILL_Week4$ gcc parser.c -o parser
amrutha@blue:~/OSSP/SKILL_Week4$ ./parser
myShell> echo Hello World

--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello
Argument 3: World
Tokenization completed successfully.
amrutha@blue:~/OSSP/SKILL_Week4$
```

Observation

The input command was successfully divided into separate tokens based on whitespace.

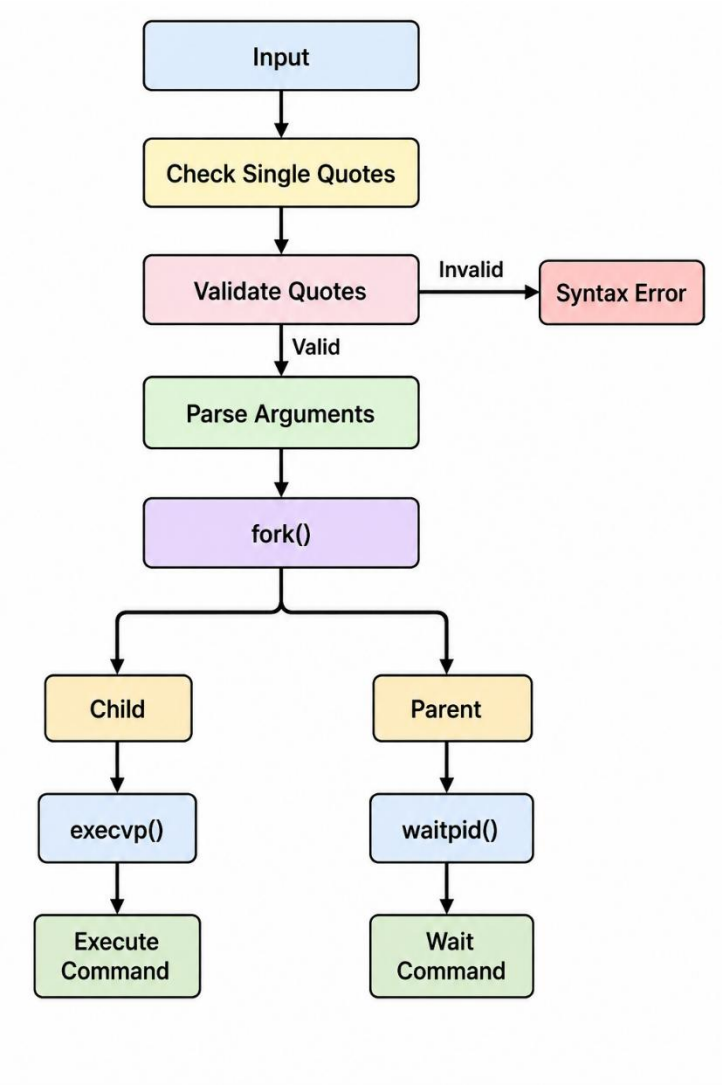
Result

The command was successfully tokenized and the individual arguments were displayed.

Task 2:
To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

Aim
To handle single quotes in shell commands and detect unmatched quotes.

Flow Chart:



Example Output:
Case1:
myShell> echo Hello

Parsing Result:
Argument 1: echo

Argument 2: Hello

Child PID: 3210
Hello

Command execution completed.

Case2:
myShell> echo 'Hello World

Syntax Error: Unmatched single quote.

Program

```
#include <stdio.h>
#include <string.h>

int main() {
    char input[200];

    printf("myShell> ");
    fgets(input, sizeof(input), stdin);

    int quotes = 0;

    for (int i = 0; input[i] != '\0'; i++) {
        if (input[i] == '\"')
            quotes++;
    }

    if (quotes % 2 != 0) {
        printf("Syntax Error: Unmatched single quote.\n");
        return 1;
    }

    printf("Parsing successful.\n");
    printf("Literal content preserved.\n");

    return 0;
}
```

Output

```
amrutha@blue:~/OSSP/SKILL_Week4$ nano single_quote.c
amrutha@blue:~/OSSP/SKILL_Week4$ gcc single_quote.c -o single_quote
amrutha@blue:~/OSSP/SKILL_Week4$ ./single_quote
myShell> echo 'Hello World'
Parsing successful.
Literal content preserved.
amrutha@blue:~/OSSP/SKILL_Week4$ ./single_quote
myShell> echo 'Hello World
Syntax Error: Unmatched single quote.
amrutha@blue:~/OSSP/SKILL_Week4$
```

Observation

Single quotes were recognized and the quoted content was preserved. An unmatched single quote was detected as a syntax error.

Result

Single-quoted strings were successfully handled and invalid quoted input was detected.

Task 3:
To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

Aim
To handle double quotes in shell commands, preserve spaces and detect unmatched double quotes.

Example Output:

Case1:
myShell> echo "Hello $USER"

--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello raghu

Child Process
PID = 3022

Hello raghu

Child process completed.

Case2:
myShell> echo "Hello World

Syntax Error: Unmatched double quote.

Program
```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char input[200];

    printf("myShell> ");
    fgets(input, sizeof(input), stdin);
```

```
int quotes = 0;

for (int i = 0; input[i] != '\0'; i++) {
    if (input[i] == '"')
        quotes++;
}

if (quotes % 2 != 0) {
    printf("Syntax Error: Unmatched double quote.\n");
    return 1;
}

printf("--- Parsed Arguments ---\n");

char *start = input;
int arg = 1;

while (*start != '\0' && *start != '\n') {
    while (*start == ' ')
        start++;

    if (*start == '\0' || *start == '\n')
        break;

    if (*start == '"') {
        start++;
        char *end = strchr(start, '"');

        if (end) {
            *end = '\0';
            printf("Argument %d: %s\n", arg++, start);
            start = end + 1;
        }
    } else {
        char *end = start;

        while (*end != ' ' && *end != '\n' && *end != '\0')
            end++;

        char temp = *end;
```

```
            *end = '\0';

            printf("Argument %d: %s\n", arg++, start);

            if (temp == '\0' || temp == '\n')
                break;

            start = end + 1;
        }
    }

    return 0;
}
```

Output

```
amrutha@blue:~/OSSP/SKILL_Week4$ nano double_quote.c
amrutha@blue:~/OSSP/SKILL_Week4$ gcc double_quote.c -o double_quote
amrutha@blue:~/OSSP/SKILL_Week4$ ./double_quote
myShell> echo "Hello World"
--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello World
amrutha@blue:~/OSSP/SKILL_Week4$ ./double_quote
myShell> echo "Hello World
Syntax Error: Unmatched double quote.
amrutha@blue:~/OSSP/SKILL_Week4$
```

Observation

Double-quoted text was treated as a single argument and spaces inside the quotes were preserved. An unmatched double quote was also detected.

Result

Double-quoted strings were successfully parsed and syntax errors were detected.

Task 4: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

Program

```
#include <stdio.h>
#include <string.h>
int main() {
    char input[200];

    printf("myShell> ");
    fgets(input, sizeof(input), stdin);

    printf("--- Parsed Input ---\n");

    for (int i = 0; input[i] != '\0' && input[i] != '\n'; i++) {
        if (input[i] == '\\' && input[i + 1] != '\0') {
            i++;
            printf("%c", input[i]);
        } else {
            printf("%c", input[i]);
        }
    }

    printf("\nParsing completed successfully.\n");

    return 0;
}
```

Output

```
amrutha@blue:~/OSSP/SKILL_Week4$ nano escape.c
amrutha@blue:~/OSSP/SKILL_Week4$ gcc escape.c -o escape
amrutha@blue:~/OSSP/SKILL_Week4$ ./escape
myShell> echo Hello\ World
--- Parsed Input ---
echo Hello World
Parsing completed successfully.
amrutha@blue:~/OSSP/SKILL_Week4$ ./escape
myShell> echo Hello\@World
--- Parsed Input ---
echo Hello@World
Parsing completed successfully.
amrutha@blue:~/OSSP/SKILL_Week4$
```

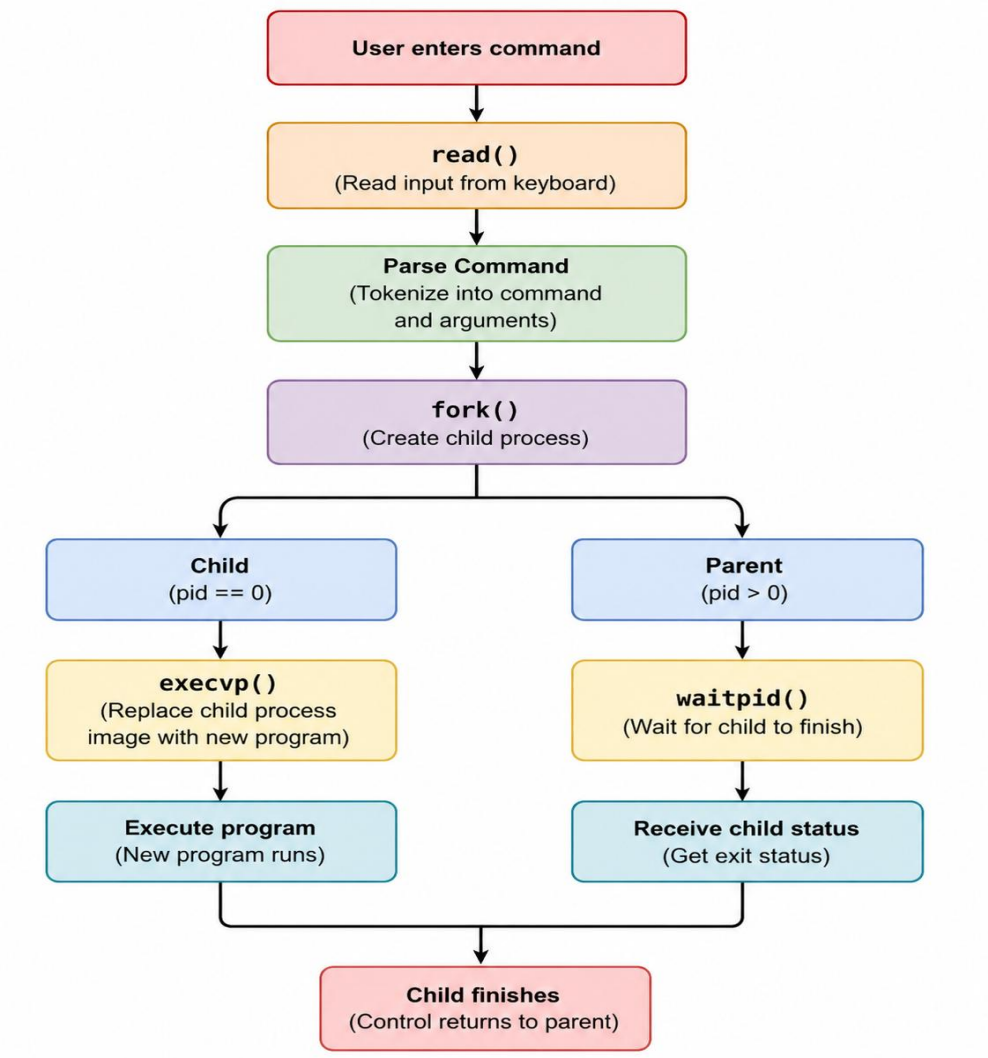
Observation
The escape character was recognized and the escaped space and special character were preserved as part of the input.

Result
Escape sequences were successfully handled and the original characters were preserved.

Task 5: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

Aim
To create a child process and execute a command using fork() and exec().

Flow Chart:



Sample Output:
Case 1:
myshell$ ls
Parent Process PID: 1200
Created Child PID: 1201
Child Process PID: 1201
file1.txt
file2.txt
Child exited with status: 0
Case 2:
myshell$ ls @l
Parent Process PID: 82
Child Process PID: 83
Created Child PID: 83
ls: cannot access '@l': No such file or directory
Child exited with status: 2

Program

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main(int argc, char *argv[]) {
   if (argc < 2) {
      printf("Usage: %s <command> [arguments...]\n", argv[0]);
      return 1;
   }
   pid_t pid = fork();
   if (pid < 0) {
      perror("fork failed");
      return 1;
   }

   if (pid == 0) {
      printf("Child Process PID: %d\n", getpid());
      execvp(argv[1], &argv[1]);

      perror("exec failed");
      exit(1);
```

```
    }
    else {
        int status;

        printf("Parent Process PID: %d\n", getpid());
        printf("Created Child PID: %d\n", pid);
        waitpid(pid, &status, 0);

        if (WIFEXITED(status))
            printf("Child exited with status: %d\n",
                WEXITSTATUS(status));
    }
    return 0;
}
```

Output

```
amrutha@blue:~/OSSP/SKILL_Week4$ nano execute.c
amrutha@blue:~/OSSP/SKILL_Week4$ gcc execute.c -o execute
amrutha@blue:~/OSSP/SKILL_Week4$ ./execute ls
Parent Process PID: 1115
Created Child PID: 1116
Child Process PID: 1116
double_quote    escape    execute    parser    single_quote
double_quote.c  escape.c  execute.c  parser.c  single_quote.c
Child exited with status: 0
```

```
amrutha@blue:~/OSSP/SKILL_Week4$ ./execute ls @l
Parent Process PID: 1144
Created Child PID: 1145
Child Process PID: 1145
ls: cannot access '@l': No such file or directory
Child exited with status: 2
amrutha@blue:~/OSSP/SKILL_Week4$
```

Observation

The parent process successfully created a child process, and the child executed the given command. An invalid command argument produced an error and a non-zero exit status.

Result

The command was successfully executed by the child process using exec(), and the parent process waited for its completion.